

Comparison-Derived Reward Shaping Through a Learning-Augmented Lens

Q Learning-Augmented Algorithms: Guarantees, Construction Mechanisms, and System-Level Implications [arXiv:2609.04787](#)

P MARS-RA: Rank Aggregation for Credit Assignment via Multimodal Comparisons in Embodied Multi-Agent Cooperation [arXiv:2607.27967](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE

The pair

Q surveys learning-augmented algorithms, separating prediction error, decision guarantees, and the cost of obtaining predictions. P uses multimodal pairwise comparisons and Bradley–Terry rank aggregation to construct potential-based rewards for cooperative multi-agent learning.

The connexion pursued

The initial question was whether Q’s consistency, robustness, and smoothness framework could characterize P’s response to inaccurate comparisons. The peer discussion found that this mapping is mostly a matter of terminology for ideal policy quality, but it exposes a testable question about finite-training benefit per comparison query [\[ledger 1, 3, 6, 12, 14\]](#).

What was found

P’s score-estimation proposition concerns error relative to an assumed latent LMM preference vector. It does not bound error relative to task contribution or the quality of a learned policy; its Shapley interpretation assumes that the latent preferences already equal Shapley values [\[ledger 2, 3, 8\]](#). P’s measured comparison accuracy instead uses agreement with dense task rewards, another target whose equivalence to Shapley values is unproved [\[ledger 8\]](#).

For a trajectory with terminal potential $\psi_T = 0$, P’s shaping rule gives the pathwise identity

$$\sum_{t=0}^{T-1} \gamma^t [r_t + \rho (\gamma \psi_{t+1} - \psi_t)] = \sum_{t=0}^{T-1} \gamma^t r_t - \rho \psi_0.$$

At a fixed initial condition, arbitrary comparison outcomes therefore do not produce a useful error-dependent curve for the optimal policy’s return under the stated boundary assumptions [\[ledger 3, 9\]](#). The relevant empirical question is how shaping changes finite training.

A one-step policy-gradient reduction makes the choice of error measure precise. If comparisons produce a pre-action baseline b_i , independent of agent i 's sampled action conditional on its history, then $g_i(R - b_i)$, with $g_i = \nabla \log \pi_i(a_i | h_i)$, has unchanged expected gradient. Its conditional second moment is minimized by

$$b_i^* = \frac{\mathbb{E}[R \|g_i\|^2 | h_i]}{\mathbb{E}[\|g_i\|^2 | h_i]}.$$

The excess second moment is $\mathbb{E}[\|g_i\|^2 | h_i] \mathbb{E}[(b_i - b_i^*)^2 | h_i]$ when the baseline randomness is conditionally action-independent [ledger 5, 6]. Thus recovering a comparison ranking need not reduce gradient noise. In the ledger's two-agent Bernoulli example, the correct Shapley order yields baseline 0.711 and excess second moment about 0.0336; reversing that order yields 0.289 and about 0.0032, because the variance-optimal baseline is 0.1 [ledger 7, 9]. This example is a reduction, not a result for P's MAPPO implementation.

For MAPPO with generalized advantage estimation, there is a stronger control. Write the shaped critic explicitly as $V_t^\psi = U_t - \rho\psi_t$, and reuse the same realized potentials in rewards and critic offsets. Then, for any parameters of U ,

$$\delta_t^\psi = r_t + \rho(\gamma\psi_{t+1} - \psi_t) + \gamma V_{t+1}^\psi - V_t^\psi = r_t + \gamma U_{t+1} - U_t.$$

With matching rollout bootstraps and value targets, advantages and value-loss residuals also match the unshaped run. Paired updates should therefore agree apart from comparison-query cost, subject to consistent agent coordinates and trajectory boundaries [ledger 16, 18, 19]. P reports 30 hours for MARS-RA versus 16 hours for MAPPO at 50 million environment steps, and a substantial dependence on shaping weight, but those observations do not identify the mechanism producing the gain [ledger 5, 16, 19].

Objections and their status

Connectedness alone does not ensure P's stated finite-error bound for its unregularized Bradley-Terry maximum-likelihood estimator. With three equal-score agents and two independent fair judgments per pair, the graph is complete, yet the probability that some agent wins every judgment involving it is $3/16 > 1/9$, exceeding the proposition's failure allowance n^{-2} ; the fitted score can then diverge [ledger 4]. The proof sketch invokes regularized estimation, while P defines an unregularized estimator. This estimator issue is unresolved in the supplied material and is distinct from the proposed training question.

Repeated comparisons may reduce sampling error around the LMM's latent preference, but cannot remove a systematic gap between that preference and task contribution [ledger 8]. The one-step baseline result also does not establish MAPPO sample complexity. P does not specify enough about potential reuse, critic representation, changing active coalitions, and truncation to determine whether the cancellation identity holds in its implementation [ledger 12, 18, 19].

Outside sources and remaining question

The peers checked prior work on optimal policy-gradient baselines, including <https://jmlr.csail.mit.edu/papers/volume5/greensmith04a/greensmith04a.pdf> and <https://arxiv.org/abs/2108.08612>; potential-based difference rewards in multi-agent learning, <https://proceedings.mlr.press/v162/li22w/li22w.pdf>; and active preference queries,

<https://arxiv.org/abs/2506.09508> [ledger 10, 11]. Accordingly, neither baseline optimality nor shaping invariance is a new contribution here.

The material supports a research question, not a publication result: can a comparison-specific query and aggregation rule improve finite-training progress enough to justify its cost, and through which critic or optimization effect [ledger 12, 14, 19]? The smallest discriminating next step is a paired MAPPO experiment with an explicit-offset critic, identical stored potentials and bootstraps, and measurements of GAE updates, success, query count, and equal-wall-clock performance against standard MARS-RA and no shaping [ledger 16, 18, 19]. If the offset control cancels the gain, the next analysis should locate how the unconstrained critic uses the potential. No such experiment or implementation audit appears in the supplied record.